CANDOR: SCORE FOLLOWING AS SEQUENTIAL SELECTION OVER NOTE HYPOTHESES

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ABSTRACT

Sheet-image score following, tracking a live performance directly on a page of printed music, is normally posed as regression: a network reads audio and an image and produces a coordinate. We pose it instead as *selection*. A page affords a finite set of places the music could be, one per notehead, and a conditional detector already proposes them; following the score means choosing among these candidates at every instant so that the choices form a plausible trajectory. The reformulation is testable, and one measurement motivates it: on real room-microphone piano recordings a correct candidate is present at every scored onset and a causal oracle over the candidates reaches 99.6%, where reporting the most confident one reaches 80.0%. CANDOR realizes the formulation with a parameter-free transition prior and a 23k-parameter permutation-invariant scorer trained listwise on synthesized audio. It tracks 94.4% of onsets within 0.5 s of real recordings at 1.25 ms per instant on one CPU thread, and because selection is separable from proposal, one trained selector improves every detector we attach it to, by 20.3, 11.0 and 7.9 points. Three findings follow: the image features the selector reads are irrelevant while its temporal history is essential, accuracy on synthesized validation audio barely ranks selectors for real audio ($r = 0.16$ over 48 checkpoints), and the forward-motion prior that makes selection work is what makes score discontinuities hard.

Index Terms— Score following, sheet music, real-time alignment, set prediction, music information retrieval

1. INTRODUCTION

Score following, the real-time alignment of a live performance to its score, underlies automatic page turning [1] and computer accompaniment [2]. Classical systems align audio to a *symbolic* score and infer the position over a discrete set of score states [3, 4, 5]. Most music, however, exists only as printed or scanned sheets, and optical music recognition is not reliable enough to produce that symbolic score automatically [6]. A decade of work therefore follows the performance directly on the sheet *image*: the input is a page of pixels and a stream of audio, the output is where on that page the musician is.

Every such system is trained on audio synthesized from the score, and every one of them loses a large part of its accuracy on real recordings. This is the field's main open problem, and the proposed remedies have all been changes to perception: reverberant augmentation, extra proprietary recordings, or a new architecture trained end to end [7, 8, 9].

We take a different route, and it begins with a change of formulation. These systems regress: a network produces a coordinate. But a page of music affords a finite, discrete set of places the performance could be, one per notehead, and a conditional detector already proposes them. Following the score is then *selection*, choosing among those candidates at every instant subject to the choices forming a plausible trajectory, which is what symbolic followers have always done over score states. The obstacle on a sheet image had been that the state set is not given; our observation is that the detector emits it and then throws it away, keeping only its most confident box.

One measurement shows what that discards. Keeping the 256 most confident candidates per instant, we find that on real room-microphone piano recordings a candidate within the 0.5 s tolerance is present at every one of 4,149 scored onsets, and that an oracle which must commit one step at a time and may never move backwards tracks 99.6% of them. The detector's own rule, take the most confident, tracks 80.0% (Fig. 1). Nineteen of the twenty points lost to real audio are lost in the decision, not in the perception. Under a regression view this is invisible: there is no set over which to take an oracle.

CANDOR is a score follower built on this formulation, made of a candidate generator, a transition prior with no trained parameters, and a small permutation-invariant scorer that judges each candidate against the others in its instant. Our contributions are: (i) the formulation, with a candidate-set oracle that locates the synthetic-to-real gap in the decision; (ii) CANDOR, which reaches 94.4% of onsets within 0.5 s on real recordings at 1.25 ms per instant and, because selection is separable from proposal, transfers unchanged across candidate generators; (iii) three findings the formulation exposes and regression hides, on what selection needs, on how badly synthetic validation ranks models, and on where the temporal prior fails.

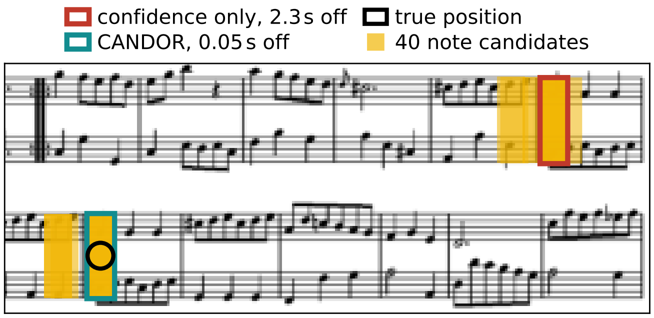

Fig. 1. One instant of a real room-microphone recording. Yellow: 40 of the note candidates the detector proposes at this instant. Taking the most confident (red) lands on the wrong staff; CANDOR (teal) selects a lower-confidence candidate on the true note (ring). The right answer was proposed and not chosen, which is the typical failure on real audio.

2. RELATED WORK

Three families. *Dense localizers* treat the page as a canvas: MM-Loc [10] and CUNet [11] emit an audio-conditioned heatmap over the whole image and read a coordinate off its centre of mass. *Agents* treat it as control: a reinforcement-learning policy [12] moves a reading head and is rewarded for staying aligned. *Conditional detectors* treat it as a query: CYOLO [7] encodes the recent audio into a vector and runs object detection so that the audio asks *which part of this page do I sound like*, and CYOLO-SB [8] adds coarse heads that also predict the enclosing system and bar. All three regress a coordinate.

One shared backbone. Beneath that difference the field has converged on a single perception module. In each case a convolutional network reads the score page and its feature maps are modulated by feature-wise linear modulation [13], $f_{\mathrm{FiLM}}(x) = s(z) \odot x + t(z)$, from a vector z encoding the recent audio; CYOLO places FiLM in its downscale and upscale blocks, and the concurrent CODA [9] conditions a CNN backbone with FiLM at its deeper stages in the same way. This is why we do not treat our backbone as a design choice. We adopt the module the field already agrees on, take its output as given, and ask what can be recovered from what it emits. A result obtained this way speaks to the family rather than to one checkpoint, and Sec. 5.1 confirms it holds across three of them.

Selection elsewhere. CODA [9] also abandons regression, and the contrast sharpens ours: it selects over *given* structure, cascading system then bar then note over the layout regions of an annotated score, with a network trained end to end and layout annotations required at test time. We select over *proposed* structure, the note candidates a frozen detector already emits, so nothing is retrained and no annotation is read. Re-scoring detections against one another has precedent in vision [14]; our scorer is a Deep-Sets [15] ranker.

3. SELECTION OVER NOTE CANDIDATES

3.1. The candidate set

A performance is processed in instants $t = 1, \ldots, T$. At each instant the generator reads the audio so far and the current page and emits note boxes $b_i = (x_i, y_i, w_i, h_i)$ with confidence o_i; we keep the $K = 256$ most confident, $\mathcal{C}_t$. Staves are concatenated along one horizontal axis, the unrolled coordinate u_i, so positions on either side of a line break are adjacent. The released CYOLO detectors emit 8,112 boxes per instant from a 52×52 grid at stride 8 and report

$$\hat{\imath}_t = \arg\max_{i \in \mathcal{C}_t} o_i, \tag{1}$$

which is the degenerate selector. CANDOR reads $\mathcal{C}_t$ and its own previous choice $\hat{u}_{t-1}$, and nothing else; no detector weight is updated.

3.2. Transition prior

Selections must form a plausible trajectory. Between instants Δ_t frames apart the performer advances by a roughly predictable amount, so we score the displacement $d_i = u_i - \hat{u}_{t-1}$ with a floored Gaussian whose mean grows with elapsed time,

$$\log \pi_i = \max\Big(-\frac{(d_i - \mu\rho_t)^2}{2\sigma^2},\ J\Big), \tag{2}$$

$\rho_t = \mathrm{clip}(\Delta_t/\Delta_0,\ 0.2,\ 8)$ and $\Delta_0 = 5$ frames, the mean inter-onset gap in training. The floor J leaves jumps possible at bounded cost, and none of (μ, σ, J) is learned. Combined with confidence, $h_i = \log o_i + \log \pi_i$ is already a competent selector.

3.3. Set scorer

The prior treats candidates independently given the history; what it cannot express is comparative, whether *this* box, among those on offer now, looks like a real notehead. Each candidate is described by $\phi_i \in \mathbb{R}^{33}$: confidence relative to the instant's best, box geometry, motion relative to the history, relation to any coarse regions the generator emits, and a neighbourhood group (how many candidates lie within 25 and 100 px, spacing to the nearest on each side, rank in reading order, size against the instant's median). We also read f_i, the 128-dimensional FiLM-conditioned feature from which the detection head scored the box, so the selector sees the same representation the generator used. A permutation-invariant network embeds the set and pools a context every candidate sees:

$$e_i = g\big([\phi_i;\ \mathrm{ReLU}(W f_i)]\big), \quad c = \Big[\tfrac{1}{K}\textstyle\sum_j e_j;\ \max_j e_j\Big], \tag{3}$$

$$s_i = r\big([e_i;\ c]\big), \tag{4}$$

with g, r two-layer perceptrons (64, 32 and 64, 1) and $W \in \mathbb{R}^{64\times128}$. The maximum is per dimension, so each candidate is judged against its competitors. The scorer has 23,203 parameters, 1.6% of the detector.

3.4. Training and decoding

Selection is supervised listwise. With ε_i the error in frames of candidate i, the target is $q_i \propto \exp(-\varepsilon_i/\tau)$, $\tau = 3$, and

$$\mathcal{L} = -\sum_{i\in\mathcal{C}_t} q_i \, \log \frac{\exp s_i}{\sum_j \exp s_j}, \tag{5}$$

so the scorer ranks a set rather than regressing a coordinate. History features come from previous selections, perturbed on 30% of samples by Laplace noise of scale 30 px, which exposes training to the drift of sequence prediction [16]. At each instant CANDOR selects

$$\hat{\imath}_t = \arg\max_{i\in\mathcal{C}_t} \; \beta\, s_i + (1-\beta)\, h_i, \tag{6}$$

causally and greedily: one forward pass over 256 candidates, 1.25 ms on one CPU thread. Training on the residual $s_i - h_i$ and selecting $\arg\max_i(s_i + h_i)$ is Eq. 6 at $\beta = 0.5$ and removes all four scalars.

4. EXPERIMENTAL SETUP

Data. Selectors are trained on 353 MSMD pieces [17], synthesized and convolved with measured room impulse responses [18], and validated on 19. The evaluation set is 16 pieces, 25 pages and 4,149 onsets played on a Yamaha AvantGrand N2 and captured with a room microphone [12]; a held-out set of 80 further pieces is rendered with a real impulse response and noise from 12 to 0.5 dB SNR. We train with AdamW ($3 \cdot 10^{-3}$, cosine, 40 epochs, batch 256).

Metric. The ratio of onsets whose selected position maps to within 0.05 to 5 s of the true time, 0.5 s primary, with 95% intervals bootstrapped over pieces [19].

The four scalars (μ, σ, J, β) are fitted leave-one-piece-out: for each evaluation piece they are chosen on the other 15, pieces weighted equally, so none influences its own decoding. Every CANDOR number is cross-validated this way and the residual variant needs no fit at all. Seeds move results by about a point, so rows are seed means with deviations.

5. RESULTS

5.1. Selection is separable from proposal

Table 1 places CANDOR against the literature and Table 2 decomposes it. The same selector, retrained only on whichever candidates the generator emits, holds across three generators of very different quality. It is worth 20.3 points on the weakest, whose training data is entirely public, lifting it past where the strongest generator begins, and reaches 94.4% on the strongest. A selector does not need the proposals to be good; it needs them to contain the answer, which the oracle says they do.

The tightest thresholds qualify this. CANDOR over CYOLO-SB candidates reaches .690 and .714 at 0.05 and 0.1 s, below CODA's .721 and .743, because a selector can only be as precise as the boxes it is given and that generator's are coarser; over CYOLO and CYOLO-SB+A candidates it is ahead at every threshold. Selection recovers which notehead, not where within it.

Two rows keep the claim honest. The parameter-free prior alone accounts for roughly half the gain, so the learned component is not carrying the result by itself, and the residual variant, which fits no scalar to any evaluation data, reaches 88.7%. Selection is also cheap: 23k parameters and 1.25 ms per instant against 12.8 ms on a GPU for the closest comparable system [9].

5.2. What selection needs

Not the image features. Feeding the selector the FiLM-conditioned backbone feature, DINOv2 patch features [20] at the same grid, or no image feature at all gives 90.6, 90.9 and 90.9, a spread well inside the 0.9-point seed deviation. On synthesized validation audio the same three differ by 3.4 points, so the features are learnable and useful there and worth nothing where it counts. Replacing the generator's audio encoder is no better: a MERT tower [21] costs more than 40 points, and Mamba pending. Perception is not the binding constraint, which is what the oracle predicted.

But the history. Removing the features computed against the previous selection costs 7.2 points, the largest single ablation here. Teacher forcing attributes about two thirds of the remaining error to drift: given the true previous position the same selector reaches 98.0% against the causal oracle's 99.6% (Fig. 2a). Selection is sequential, and it is the sequence that is hard.

Synthetic accuracy does not rank selectors. Across 48 trained checkpoints, accuracy on synthesized validation audio correlates $r = 0.16$ with accuracy on the real recordings (Fig. 2b); the best-validation checkpoint scores 89.3 on real audio, below a selector reading no image features at all. A reverberant held-out set does better ($r = 0.80$) and still mispicks. This is why the four scalars are fitted on the target recordings under cross-validation rather than on a synthetic proxy, and it is a caution for the field.

5.3. Where the temporal prior fails

Eq. 2 assumes forward motion, which is why selection works and also why a repeat or a jump breaks it. Following the protocol of [9] we splice three discontinuities into each performance at random destinations, re-running the generator on the spliced audio so its conditioning is genuinely wrong afterwards, and

Table 1. Real piano recordings, 16 pieces and 4,149 onsets captured with a room microphone. Ratio of onsets tracked within each error threshold. † generator trained with additional proprietary recordings, public checkpoint. *Layout*: needs system and bar annotations at test time. *Trains*: what is fitted for the task. Published figures as reported in [9].

	Method	Onset error threshold (s)					Layout	Trains
		$\leq$0.05	$\leq$0.1	$\leq$0.5	$\leq$1.0	$\leq$5.0		
ours	CANDOR over CYOLO candidates	.726	.747	.916	.950	.991	no	23k selector
	CANDOR over CYOLO-SB candidates	.690	.714	.906	.942	.982	no	23k selector
	CANDOR over CYOLO-SB+A† candidates	**.760**	**.785**	**.944**	**.969**	**.996**	no	23k selector
prior work	MM-Loc [10], dense	.364	.406	.585	.611	.735	no	1.2M end to end
	RL agent [12], control	.185	.200	.476	.603	.901	no	policy
	CUNet [11], dense	.113	.125	.224	.266	.443	no	4.5M end to end
	CYOLO [7], query	.563	.581	.712	.749	.919	no	1.5M detector
	CYOLO-SB [8], query	.610	.630	.799	.832	.960	no	1.5M detector
	CYOLO-SB+A† [8]	.682	.706	.865	.891	.981	no	1.5M detector
	CODA [9], given structure	.721	.743	.883	.914	.987	yes	2.0M end to end

Table 2. Components of CANDOR, cross-validated per piece on the real recordings at 0.5 s. *argmax* is Eq. 1 over the same candidates, *prior* adds Eq. 2, *full* adds the scorer.

	n	argmax	prior	full
Candidate generator				
CYOLO [7]	3	71.3	72.5	91.6 ± 1.4
CYOLO-SB [8]	6	80.0	86.5	90.6 ± 0.9
CYOLO-SB+A†	3	86.5	90.8	$\mathbf{94.4 \pm 0.9}$
Selector, on CYOLO-SB candidates				
residual, no fitted scalars	3	80.0	86.5	88.7 ± 0.9
without temporal history	3	80.0	86.5	83.7 ± 0.4
Image features f_i the selector reads				
FiLM-conditioned backbone (default)	6	80.0	86.5	90.6 ± 0.9
DINOv2 [20]	1	80.0	86.5	90.9
none	1	80.0	86.5	90.9
Audio encoder inside the generator, retrained				
MERT [21]	3	33.4	31.9	46.2 ± 9.1
Mamba [22]	3	.	.	.

excluding jumps across a page boundary where our state resets anyway, leaving 41.

Table 3 reports a failure. CANDOR is worse after a jump than taking the most confident candidate, 0.15 against 0.22 of onsets tracked over the next five seconds, because the prior pins the trajectory to a stale position. Suppressing it on detected silence recovers part of that when a break exists and nothing when it does not; triggering instead on the selector's own confidence needs no break and works in both cases, but only returns to the confidence baseline. A prior worth eleven points on continuous playing is strong enough to be wrong at a discontinuity, and handling both in one selector is open.

Table 3. Recovery from 41 spliced discontinuities. *rec@1*: fraction first tracked correctly within 1 s. *post*: onsets tracked within 1 s over the following 5 s.

Selector	break present		no break	
	rec@1	post	rec@1	post
confidence only (1)	0.32	0.22	0.27	0.22
CANDOR	0.12	0.15	0.10	0.15
+ silence trigger	0.27	0.19	0.10	0.15
+ confidence trigger	0.15	0.22	0.20	**0.23**

5.4. Robustness and budget

Under a real impulse response and noise on 80 held-out pieces the scorer beats the prior at every level from 12 to 0.5 dB, by less as the SNR falls. On 94 synthetic test pieces, where confidence alone selects .893 at 0.5 s, CANDOR reaches .955. Halving the budget to $K = 128$ costs 0.5 points and the correct candidate sits at rank 27 at the 90th percentile: the tail is load-bearing.

6. CONCLUSION

Posing sheet-image score following as selection over note candidates makes the synthetic-to-real gap measurable and mostly closable: the answer is already proposed, and a 23k-parameter selector finds it 94.4% of the time at 1.25 ms per instant, transferring across generators because selection and proposal are separate concerns. What it needs is its own history, not better pixels, and the constraint it relies on is what fails at a discontinuity. The correct page is given, all performances are piano, and 16 pieces cannot resolve a point or two. Pitch agreement is the natural next history-independent cue.

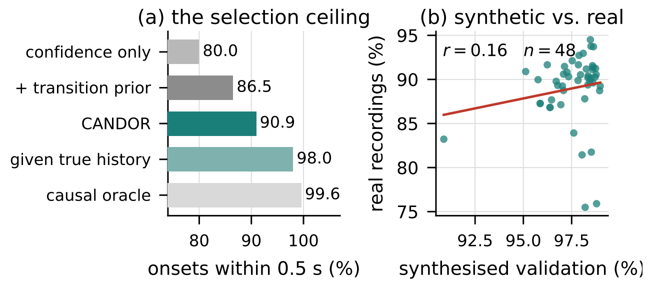

Fig. 2. (a) Every step from the degenerate selector to the ceiling on the real recordings. CANDOR to *given true history* is drift; above it is ranking. (b) Each point is one of 48 trained selectors: accuracy on synthesised validation audio against accuracy on the real recordings.

7. COMPLIANCE WITH ETHICAL STANDARDS

This is a numerical study that uses only publicly available data; no ethical approval was required.

8. ACKNOWLEDGMENTS

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9. REFERENCES

[1] Andreas Arzt, Gerhard Widmer, and Simon Dixon, “Automatic page turning for musicians via real-time machine listening,” in *Proc. European Conference on Artificial Intelligence (ECAI)*, 2008, pp. 241–245.

[2] Roger B. Dannenberg, “An on-line algorithm for real-time accompaniment,” in *Proc. International Computer Music Conference (ICMC)*, 1984.

[3] Arshia Cont, “A coupled duration-focused architecture for real-time music-to-score alignment,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 32, no. 6, pp. 974–987, 2010.

[4] Eita Nakamura, Philippe Cuvillier, Arshia Cont, Nobutaka Ono, and Shigeki Sagayama, “Autoregressive hidden semi-Markov model of symbolic music performance for score following,” in *Proc. International Society for Music Information Retrieval Conference (ISMIR)*, 2015, pp. 392–398.

[5] Simon Dixon and Gerhard Widmer, “MATCH: A music alignment tool chest,” in *Proc. International Society for Music Information Retrieval Conference (ISMIR)*, 2005.

[6] Jorge Calvo-Zaragoza, Jan Hajič jr., and Alexander Pacha, “Understanding optical music recognition,” *ACM Computing Surveys*, vol. 53, no. 4, 2020.

[7] Florian Henkel and Gerhard Widmer, “Multi-modal conditional bounding box regression for music score following,” in *Proc. European Signal Processing Conference (EUSIPCO)*, 2021.

[8] Florian Henkel and Gerhard Widmer, “Real-time music following in score sheet images via multi-resolution prediction,” *Frontiers in Computer Science*, vol. 3, pp. 718340, 2021.

[9] Yining Yang, Ruogu Chen, and Jie Han, “CODA: Cascaded online discontinuity-aware alignment for real-time image-based score following,” in *Proc. International Society for Music Information Retrieval Conference (ISMIR)*, 2026.

[10] Matthias Dorfer, Andreas Arzt, and Gerhard Widmer, “Towards score following in sheet music images,” in *Proc. International Society for Music Information Retrieval Conference (ISMIR)*, 2016.

[11] Florian Henkel, Rainer Kelz, and Gerhard Widmer, “Learning to read and follow music in complete score sheet images,” in *Proc. International Society for Music Information Retrieval Conference (ISMIR)*, 2020.

[12] Florian Henkel, Stefan Balke, Matthias Dorfer, and Gerhard Widmer, “Score following as a multi-modal reinforcement learning problem,” *Transactions of the International Society for Music Information Retrieval*, vol. 2, no. 1, 2019.

[13] Ethan Perez, Florian Strub, Harm de Vries, Vincent Dumoulin, and Aaron Courville, “FiLM: Visual reasoning with a general conditioning layer,” in *Proc. AAAI Conference on Artificial Intelligence*, 2018.

[14] Jan Hosang, Rodrigo Benenson, and Bernt Schiele, “Learning non-maximum suppression,” in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.

[15] Manzil Zaheer, Satwik Kottur, Siamak Ravanbakhsh, Barnabás Póczos, Ruslan Salakhutdinov, and Alexander Smola, “Deep sets,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[16] Samy Bengio, Oriol Vinyals, Navdeep Jaitly, and Noam Shazeer, “Scheduled sampling for sequence prediction with recurrent neural networks,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2015.

[17] Matthias Dorfer, Jan Hajič jr., Andreas Arzt, Harald Frostel, and Gerhard Widmer, “Learning audio-sheet music correspondences for cross-modal retrieval and piece identification,” *Transactions of the International Society for Music Information Retrieval*, vol. 1, no. 1, 2018.

[18] James Traer and Josh H. McDermott, “Statistics of natural reverberation enable perceptual separation of sound and space,” *Proceedings of the National Academy of Sciences*, vol. 113, no. 48, pp. E7856–E7865, 2016.

[19] Bradley Efron and Robert J. Tibshirani, *An Introduction to the Bootstrap*, Chapman & Hall, 1993.

[20] Maxime Oquab, Timothée Darcet, Théo Moutakanni, et al., “DINOv2: Learning robust visual features without supervision,” *Transactions on Machine Learning Research*, 2024.

[21] Yizhi Li, Ruibin Yuan, Ge Zhang, et al., “MERT: Acoustic music understanding model with large-scale self-supervised training,” in *Proc. International Conference on Learning Representations (ICLR)*, 2024.

[22] Albert Gu and Tri Dao, “Mamba: Linear-time sequence modeling with selective state spaces,” in *Proc. Conference on Language Modeling (COLM)*, 2024.